POINTS TO BE REMENBER- 1) Har question ke baad ek example dena nhi to omesh marks katega.

2) Iske alawa kuch padhna hoga to MDN ek website hai nhi to firr W3 schools hai …use refer karna.

3) Maine kuch important question ke baad uska source diya hai to extra ka padhna hoga to uskoclick karke access kar lena.

Q1) **What is HTML?**

HTML: HyperText Markup Language

HTML (HyperText Markup Language) is the most basic building block of the Web. It defines the meaning and structure of web content. Other technologies besides HTML are generally used to describe a web page's appearance/presentation (CSS) or functionality/behavior (JavaScript).

"Hypertext" refers to links that connect web pages to one another, either within a single website or between websites. Links are a fundamental aspect of the Web. By uploading content to the Internet and linking it to pages created by other people, you become an active participant in the World Wide Web.

HTML uses "markup" to annotate text, images, and other content for display in a Web browser. HTML markup includes special "elements" such as `<head>`, `<title>`, `<body>`, `<header>`, `<footer>`, `<article>`, `<section>`, `<p>`, `<div>`, `<span>`, `<img>`, `<aside>`, `<audio>`, `<canvas>`, `<datalist>`, `<details>`, `<embed>`, `<nav>`, `<search>`, `<output>`, `<progress>`, `<video>`, `<ul>`, `<ol>`, `<li>` and many others.

An HTML element is set off from other text in a document by "tags", which consist of the element name surrounded by `<` and `>`. The name of an element inside a tag is case-insensitive. That is, it can be written in uppercase, lowercase, or a mixture. For example, the `<title>` tag can be written as `<Title>`, `<TITLE>`, or in any other way. However, the convention and recommended practice is to write tags in lowercase.

Q2) Signs of good and bad websites.

Signs of a good website:

- **Clear navigation:** Easy to find what you're looking for with well-organized menus and links.
- **Consistent design:** Consistent color schemes, fonts, and layout throughout the website.
- **Mobile-friendly:** Adapts seamlessly to different screen sizes, especially on mobile devices.
- **Fast loading times:** Pages load quickly, minimizing user frustration.
- **High-quality content:** Relevant, informative, and well-written text.
- **Accessibility features:** Designed to be usable by people with disabilities.

Signs of a bad website:

- **Confusing navigation:** Difficult to find information or navigate through the site.
- **Inconsistent design:** Different styles, colors, and fonts on different pages.
- **Slow loading times:** Takes a long time for pages to load.
- **Not mobile-friendly:** Poor formatting on mobile devices.
- **Cluttered layout:** Too much information crammed into a small space.
- **Poorly written content:** Grammatical errors, irrelevant information, or difficult to understand language.
- **Lack of accessibility features:** Not accessible to users with disabilities.
- **Outdated design:** Looks old and outdated.
- **High bounce rate:** Many users leaving the website after viewing only one page.

Q3) What is https (request and response)?

HTTPS request and response" refers to the secure communication between a web browser (client) and a web server where all data exchanged, including the request sent by the browser and the response sent back by the server, is encrypted using TLS (Transport Layer Security), ensuring privacy and security during data transmission; essentially, it's the secure version of the standard HTTP request-response cycle where sensitive information like passwords or credit card details are protected from being intercepted by third parties.

Q4) EXPLAIN CLIENT-SERVER ARCHITECTURE.

Client-server architecture is a system that uses a central server to manage requests from clients. The server then processes the requests and sends back the results.

How it works

- Clients, such as web browsers or email clients, request services or resources from the server
- The server processes the requests and gathers the necessary information
- The server generates results and sends them back to the clients

Benefits

- **Centralized management**: Consolidates control of data, applications, and security within the server
- **Scalability**: Supports efficient scaling options, enabling businesses to grow
- **Reliability**: Distributed across multiple servers, client-server architecture enhances system reliability

Applications

- Client-server architecture is popular in web applications, databases, and other network-based systems
- Examples of computer applications that use the client–server model are email, network printing, and the World Wide Web

Q5) web browsers and web servers.

The **web browser** is an application software to explore www (World Wide Web). It provides an interface between the server and the client and it requests to the server for web documents and services. It works as a compiler to render HTML which is used to design a webpage. Whenever we search for anything on the internet, the browser loads a web page written in HTML, including text, links, images, and other items such as style sheets and JavaScript functions. Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari are examples of web browsers.

A **web server** is a software or hardware that delivers web content to users over the internet. When a user visits a website, their browser sends a request to the web server, which then sends back the requested web pages.

Q6) Difference between Web Browsers and Web servers.

Feature	Web Browser	Web Server
Definition	A software application used to access websites and display web pages.	A computer or software that hosts websites and serves web pages to clients (browsers).
Function	Sends HTTP/HTTPS requests and displays responses (web pages).	Processes HTTP/HTTPS requests and sends responses (web content).
Examples	Chrome, Firefox, Edge, Safari, Opera	Apache, Nginx, IIS, LiteSpeed
User	Used by end-users to browse the internet.	Used by website owners to host web applications.
Input	URL, user interaction, form submissions, JavaScript execution.	Requests from browsers or clients (via HTTP/HTTPS).
Output	Renders and displays web pages to users.	Sends web pages, data, or error messages.
Communication	Acts as a client by requesting web pages.	Acts as a server by responding to client requests.
Storage	Stores cached pages, cookies, history, and bookmarks.	Stores website files, databases, and server configurations.

Q7) Difference between get and post methods.

Feature	GET Method	POST Method
Purpose	Used to request data from a server.	Used to send data to a server.
Data Transmission	Sends data in the **URL (query parameters)**.	Sends data in the **body** of the request.
Visibility	Data is **visible** in the URL.	Data is **hidden** in the request body.
Security	Less secure (data can be cached, stored in history, and seen in logs).	More secure (data is not exposed in the URL).
Usage	Used for **retrieving** information (e.g., search queries, fetching pages).	Used for **submitting** information (e.g., login forms, registrations).
Caching	Can be cached by browsers and stored in history.	Cannot be cached; not stored in history.
Bookmarking	Can be bookmarked since the URL contains all data.	Cannot be bookmarked as data is in the request body.
Idempotency	**Idempotent** (same request gives the same response).	**Not idempotent** (multiple requests may have different effects).
Request Length	Limited length (depends on the browser and server, typically 2048 characters).	No restriction on data length.
Examples	`GET /search?q=hello`	`POST /login` with a JSON or form payload.

Q8) What is the structure and syntax of html?

An HTML document consists of specific tags that form the skeleton of the page. Here’s a look at a basic HTML document structure.

```html
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8" />
    <meta name="viewport" content=
        "width=device-width, initial-scale=1.0" />
    <title>Structure of HTML Document</title>
</head>

<body>
    <!-- Main content of website -->
    <h1>GeeksforGeeks</h1>
    <p>A computer science portal for geeks</p>
</body>

</html>
```

1. **<!DOCTYPE HTML>:** The <!DOCTYPE html> declaration is placed at the beginning of the document. It tells the browser that the document follows HTML5 standards, ensuring consistent rendering across browsers.
2. **<html> Tag:** The <html> tag wraps the entire document, serving as the root element of an HTML page. It typically includes the **lang attribute** to specify the language of the content.
3. **<head> Section:** The <head> section contains metadata, scripts, styles, and other information not displayed directly on the page but essential for functionality and SEO.
4. **<body> Section:** The <body> section contains all the visible content of the web page, including text, images, videos, links, and more. This is where you’ll add the main elements to display on the page.

Q9)Elements in html.(click here for more)

An **HTML Element** consists of a **start tag**, **content**, and an **end tag**, which together define the element’s structure and functionality. Elements are the basic building blocks of a webpage and can represent different types of content, such as text, links, images, or headings.

For example, the <p> element for paragraphs includes opening and closing tags with text content in between.

Syntax:

```
<tagname >Your Contents... </tagname>
```

HTML Element Code Example:

In this example <p> is a starting tag, </p> is an ending tag and it contains some content between the tags, which form an element.

```
<!-- HTML code to illustrate HTML elements -->
<!DOCTYPE html>
<html>

<head>
    <title>HTML Elements</title>
</head>

<body>
    <p>Welcome to GeeksforGeeks!</p>
</body>

</html>
```

Q10) Explain text formatting tags.(click here for more)

HTML text formatting refers to the use of specific HTML tags to modify the appearance and structure of text on a webpage. It allows you to style text in different ways, such as making it bold, italic, underlined, highlighted, or struck-through.

HTML text formatting can be divided into two main categories: **Logical Tags** and **Physical Tags.**

1. **Logical Tags**

Logical tags convey the meaning or importance of the text without necessarily altering its visual appearance. These tags help browsers, search engines, and assistive technologies understand the purpose of the text.

- <em>: Emphasizes text, typically rendered in italics. It implies that the text carries special importance or requires emphasis.
- <strong>: Marks text as important, often displayed in bold. It implies the content is of strong importance.

2. **Physical Tags**

Physical tags directly affect how text looks on the webpage by changing the font, size, or style.

- [<b>]: Displays text in bold without implying importance.
- [<i>]: Italicizes text without any implied emphasis.

Q11) What is html semantics?(click here for more)

HTML5 introduced a range of **semantic elements** that clearly describe their purpose in human and machine-readable language. Unlike non-semantic elements, which provide no information about their content, semantic elements clearly define their content.
For instance, **<form>**, **<table>**, and **<article>** tags clearly define the content and purpose, to the browser.

Why Use Semantic HTML Tags?

- **Accessibility:** Semantic elements make web pages more accessible. Screen readers and other assistive technologies can interpret the structure and navigate the content more efficiently.
- **SEO:** Better structured data leads to better SEO. Search engines prioritize well-structured content that uses semantic elements correctly, as it's easier to index.
- **Maintainability:** Semantic HTML helps create a logically structured document, which is easier to read and maintain.

Semantic Elements

Here are some of the fundamental HTML5 semantic elements that you should use to structure your web content:

1. <article>
2. <aside>
3. <details>
4. <figcaption>
5. <figure>
6. <footer>
7. <header>
8. <main>
9. <mark>
10. <nav>
11. <section>

Q12) Explain html lists.(click me)

An **HTML List** allows you to organize data on web pages into an ordered or unordered format to make the information easier to read and visually appealing. HTML Lists are very helpful for creating structured, accessible content in web development.

Types of HTML Lists

There are three main types of lists in HTML:

1. **Unordered Lists (<ul>):** These lists are used for items that do not need to be in any specific order. The list items are typically marked with bullets.
2. **Ordered Lists (<ol>):** These lists are used when the order of the items is important. Each item in an ordered list is typically marked with numbers or letters.
3. **Description Lists (<dl>):** These lists are used to contain terms and their corresponding descriptions.

Ordered lists- HTML Ordered List is created by the HTML ** tag**, to display elements in an ordered form, either numerical or alphabetical. Each item within the list is placed within a <li> tag, which stands for "list item".
The list is automatically numbered by the browser, but the style of numbering can be adjusted using attributes and CSS.

```html
<!DOCTYPE html>
<html>
<head>
    <title>Simple Ordered List</title>
</head>
<body>
    <h2>My To-Do List</h2>
    <ol>
        <li>Go grocery shopping</li>
        <li>Pay utility bills</li>
        <li>Prepare dinner</li>
    </ol>
</body>
</html>
```

Type	Descriptions
type="1"	This will list the items with numbers (default)
type="A"	This will list the items in uppercase letters.
type="a"	This will list the items in lowercase letters.
type="I"	This will list the items with uppercase Roman numbers.
type="i"	This will list the items with lowercase Roman numbers.

Unordered list- An **HTML Unordered List** is defined with the ** tag**, where “ul” stands for “unordered list.” Each item within the list is marked by a <li> tag, standing for “list item.”

The items in an unordered list are typically displayed with bullet points, which can be styled or changed using CSS.

```
<ul>
    <li>Item 1</li>
    <li>Item 2</li>
</ul>
```

```
<!DOCTYPE html>
<html>

<head>
    <title>
        HTML Unordered Lists
    </title>
</head>

<body>
    <h2>HTML Unordered Lists</h2>

    <ul>
        <li>HTML</li>
        <li>CSS</li>
        <li>Javascript</li>
        <li>React</li>
    </ul>
</body>

</html>
```

Unordered Lists Style Types

Values	Descriptions
disc	This value sets the list marker to a bullet (default).
circle	This value sets the list marker to a circle.
square	This value sets the list marker to a square.
none	This value unmarks the list of items.

UNIT-2

Q13)What are hyperlinks and types of hyperlink?(learn more)

HTML Links, also known as **hyperlinks**, are defined by the **<a>** tag in HTML, which stands for "anchor." These links are essential for navigating between web pages and directing users to different sites, documents, or sections within the same page.

The basic attributes of the <a> tag include **href**, **title**, and **target**, among others.

Basic Syntax of an HTML Link:

```
<a href="https://www.example.com">Visit Example</a>
```

Note: A hyperlink can be represented by an image or any other HTML element, not just text.

A Simple HTML Link Example

In this example, we contains a paragraph instructing users to click on the link labeled "GeeksforGeeks," which directs to the website "https://www.geeksforgeeks.org".

```
<!DOCTYPE html>
<html>

<head>
    <title>HTML Links</title>
</head>

<body>
    <p>Click on the following link</p>
    <a href="https://www.geeksforgeeks.org">
        GeeksforGeeks
    </a>
</body>

</html>
```

HTML Links – Target Attribute

The `target` attribute in the `<a>` tag specifies where to open the linked document. It controls whether the link opens in the same window, a new window, or a specific frame.

Attribute	Description
`_blank`	Opens the linked document in a new window or tab.
`_self`	Opens the linked document in the same frame or window as the link. (Default behavior)
`_parent`	Opens the linked document in the parent frame.
`_top`	Opens the linked document in the full body of the window.
`framename`	Opens the linked document in a specified frame. The frame's name is specified in the attribute.

Q14) HTML MEDIA i.e. Audio and video tags.(audio video)

HTML5 audio Tag

Last Updated : 25 Nov, 2024

The <audio> tag in HTML5 is used to embed audio content on a webpage. It allows you to play audio files like MP3, OGG, or WAV directly in the browser. The <audio> element provides attributes for controlling playback, such as play, pause, and volume.

- Using the <source> element enables the specification of various audio files, allowing the browser to choose the compatible format.
- The <audio> element allows integration with JavaScript, enabling the creation of custom audio controls and behaviors.
- The 'controls' attribute provides buttons for managing audio, like play, pause, and volume adjustment.
- Any text contained between the <audio> tags is visible only on browsers that can't render the <audio> element.

```html
<!DOCTYPE html>
<html>

<body>
    <audio controls>
        <source src=
"https://media.geeksforgeeks.org/wp-content/uploads/20230524142525/gfg_offline_classes_en.mp3"
                type="audio/mp3">
        <source src=
"https://media.geeksforgeeks.org/wp-content/uploads/20220913101124/audiosample.ogg"
                type="audio/mp3">
    </audio>
</body>

</html>
```

The various attributes that can be used with the "audio" tag are listed below:

Attributes	Description
Controls	Designates what controls to display with the audio player.
Autoplay	Designates that the audio file will play immediately after it loads controls.
muted	Designates that the audio file should be muted.
src	Designates the URL of the audio file.
Loop	Designates that the audio file should continuously repeat.

HTML Video

Last Updated : 24 Dec, 2024

The <video> element in HTML is used to add video content to web pages. It supports various video formats, including MP4, WebM, and Ogg. Video and audio tags are introduced in HTML5.

Syntax:

```
<video src="" controls>   </video>
```

- The **src attribute** specifies the URL of the video file.
- The controls attribute adds default video controls (play, pause, volume, etc.).

```
<html >
<body>
    <video width="320" height="240" controls>
        <source src=
"https://media.geeksforgeeks.org/wp-content/uploads/20190616234019/Canvas.move_.mp4"
            type="video/mp4">
        Sample Vedio
    </video>
</body>
</html>
```

- The <video> tag defines the video player, with width and height attributes setting its dimensions.
- The controls attribute adds playback controls like play, pause, and volume.

HTML Video Tags

Here are the HTML tags used for adding video content:

- **<video>:** Defines a video or movie on a webpage.
- **<source>:** Specifies multiple media resources for video or audio elements (e.g., different video formats).
- Adds text tracks (like subtitles or captions) to a video or audio element.

Additional Attributes

Attributes	Description
Autoplay	Starts playing the video automatically.
Preload	Provides a hint to the browser about the best user experience.
Loop	Automatically loops the video.
height	It sets the height of the video in CSS pixels.
width	It determines the width of the video display area on the web page.
Controls	It shows the default video controls like play, pause, volume, etc.
Muted	Mutes the audio.
Poster	Displays an image preview before video loading.

Q15) HTML TABLES. (KNOW MORE)

HTML Tables

Last Updated : 05 Feb, 2025

HTML Tables allow you to arrange data into rows and columns on a web page, making it easy to display information like schedules, statistics, or other structured data in a clear format.

What is an HTML Table?

An HTML table is created using the <table> tag. Inside this tag, you use

- *<**tr**> to define table rows,*
- *<**th**> for table headers, and*
- *<**td**> for table data cells*

Each <tr> represents a row, and within each row, <th> or <td> tags represent the cells in that row, which can contain text, images, lists, or even another table.

```html
<!-- index.html -->
<!DOCTYPE html>
<html>

<body>
    <table>
        <tr>
            <th>Firstname</th>
            <th>Lastname</th>
            <th>Age</th>
        </tr>
        <tr>
            <td>Priya</td>
            <td>Sharma</td>
            <td>24</td>
        </tr>
        <tr>
            <td>Arun</td>
            <td>Singh</td>
            <td>32</td>
        </tr>
        <tr>
            <td>Sam</td>
            <td>Watson</td>
            <td>41</td>
        </tr>
    </table>
</body>

</html>
```

Firstname	Lastname	Age
Priya	Sharma	24
Arun	Singh	32
Sam	Watson	41

Tags used in HTML Tables

HTML Tags	Descriptions
<table>	Defines the structure for organizing data in rows and columns within a web page.
<tr>	Represents a **row** within an HTML table, containing individual cells.
<th>	Shows a table **header** cell that typically holds titles or headings.
<td>	Represents a standard **data** cell, holding content or data.
<caption>	Provides a title or description for the entire table.

<thead>	Defines the header section of a table, often containing column labels.
<tbody>	Represents the main content area of a table, separating it from the header or footer.
<tfoot>	Specifies the footer section of a table, typically holding summaries or totals.
<col>	Defines attributes for table **columns** that can be applied to multiple columns at once.
<colgroup>	Groups together a **set of columns** in a table to which you can apply formatting or properties collectively.

UNIT -3

Q16) Html forms.(KNOW MORE)

HTML Forms

Last Updated : 15 Oct, 2024

HTML Forms use the <form> tag to collect user input through various interactive controls. These controls range from text fields, numeric inputs, and email fields to password fields, checkboxes, radio buttons, and submit buttons.

Syntax:

```
<form>
      <!--form elements-->
</form>
```

Form Elements

The HTML <form> comprises several elements, each serving a unique purpose. For instance, the <label> element defines labels for other <form> elements. On the other hand, the <input> element is versatile and can be used to capture various types of input data such as text, password, email, and more simply by altering its type attribute.

Elements	Descriptions
<label>	It defines labels for <form> elements.
<input>	It is used to get input data from the form in various types such as text, password, email, etc by changing its type.
<button>	It defines a clickable button to control other elements or execute a functionality.
<select>	It is used to create a drop-down list.
<textarea>	It is used to get input long text content.
<fieldset>	It is used to draw a box around other form elements and group the related data.
<legend>	It defines a caption for fieldset elements
<datalist>	It is used to specify pre-defined list options for input controls.
<output>	It displays the output of performed calculations.
<option>	It is used to define options in a drop-down list.
<optgroup>	It is used to define group-related options in a drop-down list.

Q17) Input tags in html.(know more)

HTML input Tag

Last Updated : 09 Jan, 2025

The <input> tag in HTML is used to collect user input in web forms. It supports various input types such as text, password, checkboxes, radio buttons, and more.

An input field can be of various types depending upon the attribute type. The Input tag is an empty element that only contains attributes. For defining labels for the input element, <label> can be used.

Note: The <input> Tag supports the Global Attributes & the Event Attributes in HTML

Syntax

```
<input type = "value" ... />
```

```html
<html>
<body>
    <form>
        <label for="username">Username:</label>
        <input type="text" id="username" name="username">
    </form>
</body>
</html>
```

- The <label> element with the for attribute associates the label "Username:" with the <input> field, enhancing accessibility.
- The <input> element of type "text" creates a single-line text input field where users can enter their username.

Q18) EMOJIS TAGS. (KNOW MORE)

As we know, Unicode is a system that assigns a unique number (code point) to every character, symbol, or emoji, making it possible for different devices and platforms to display the same character consistently.

```
Emojis are just special Unicode characters displayed as images.
```

In HTML, emojis are typically included using their Unicode escape sequences or directly as characters. Emojis are usually rendered as images by browsers, but in the backend, they are just text characters with unique Unicode code points.

Adding Emojis in HTML Code

```html
<!DOCTYPE html>
<html lang="en">

<head>
    <title>UTF-8 Character Example</title>
</head>

<body>
    <!-- Direct Putting Emoji as a character -->
    <p>Here is a smiley face: 😊</p>

    <!-- Using unicode Escape -->
    <p>Here is a smiley face: &#128522;</p>
</body>

</html>
```

HTML Emojis Examples

1. HTML Emojis using Unicode Decimal reference

HTML Emojis are represented using Unicode decimal references like 😄 to display emojis, enabling the rendering of various symbols and icons in web content.

Example 1: Represent the following emoji in a webpage.

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <title>HTML Emojis Example</title>
</head>

<body>
    <h1>HTML Emojis Example</h1>
    <p>
        &#128516; This is a smiling face with open mouth
        and smiling eyes emoji
    </p>
    <p>&#9996; This is a victory hand emoji</p>
    <p>&#8986; This is a watch emoji</p>
</body>

</html>
```